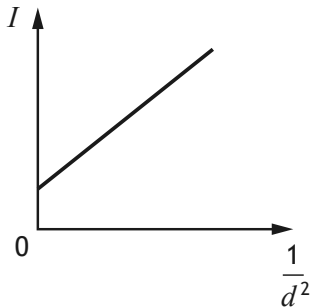


14. A student carries out an experiment to investigate how irradiance of light varies with distance.

A small lamp is placed at a distance d from a light meter. The irradiance I at this distance is displayed on the light meter. This measurement is repeated for a range of different distances.

The student uses these results to plot the graph shown.



The graph indicates that there is a systematic uncertainty in the experiment.

Which of the following alterations would be most likely to reduce the systematic uncertainty in this experiment?

- A Repeating the experiment in a darkened room
- B Repeating the readings at each distance and calculating averages
- C Decreasing the brightness of the lamp
- D Replacing the small lamp with a larger lamp
- E Increasing the range of distances